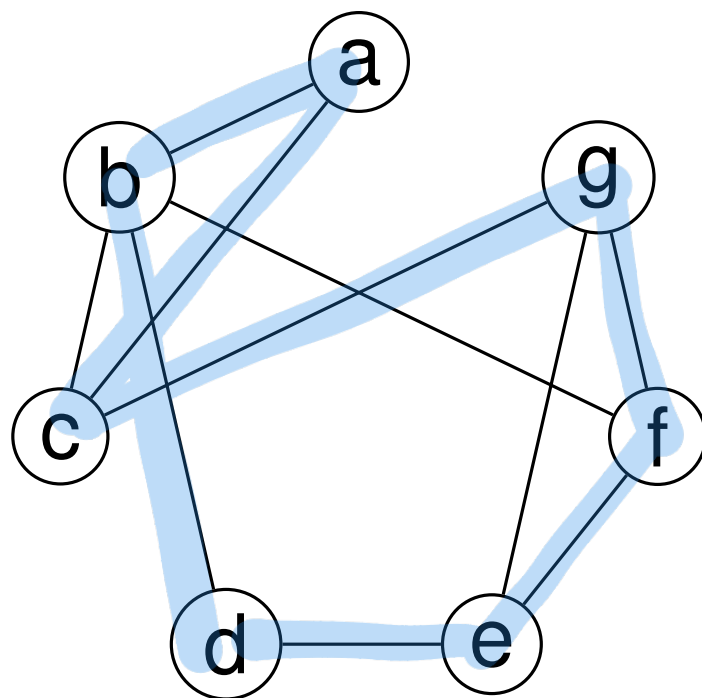


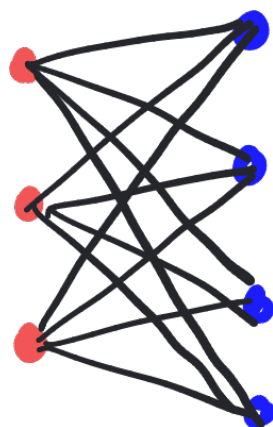
Lecture Notes: Graph Theory—Three Hamiltonian Circuit Algorithms

Math F113X Math and Society

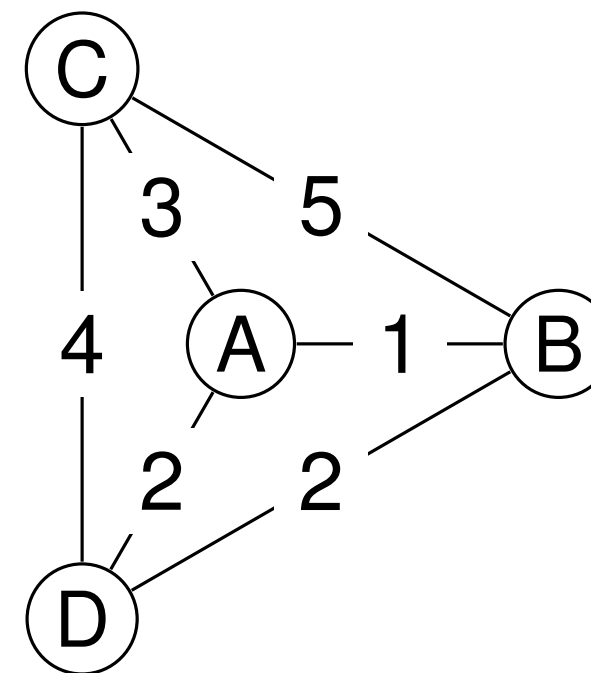
Two Distinct Questions



Can we find a Hamiltonian circuit, at all?

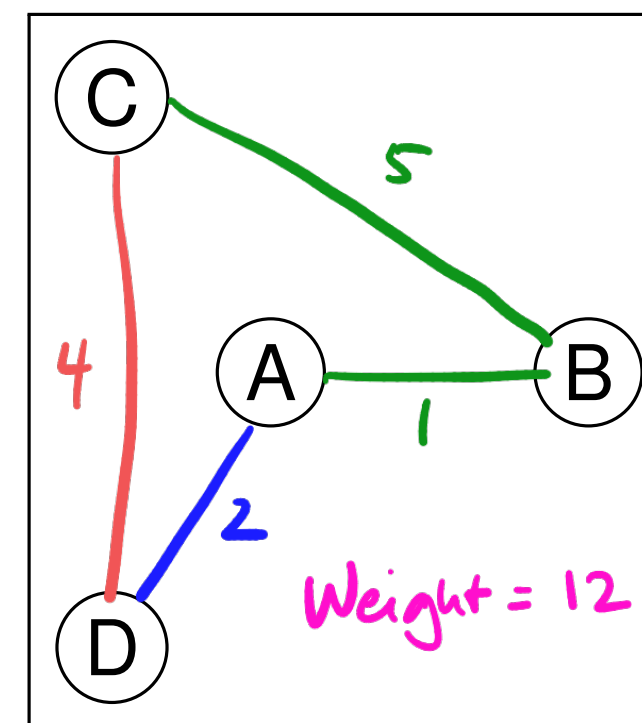
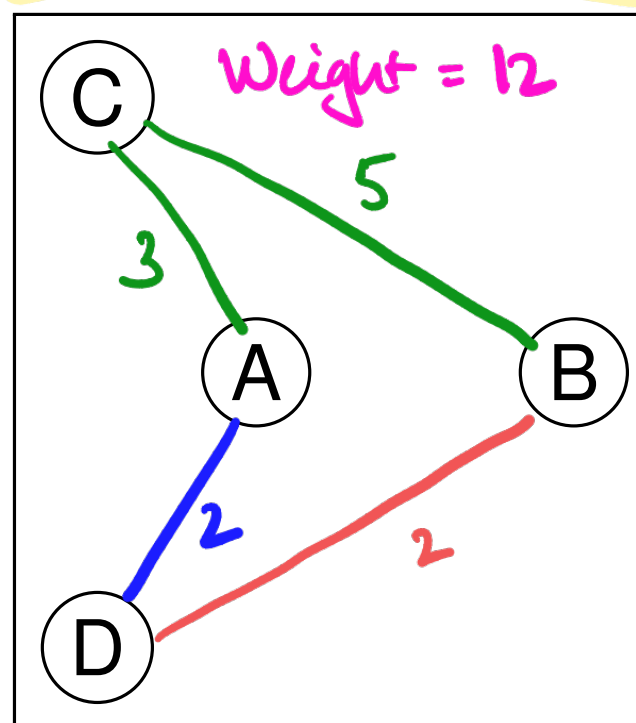
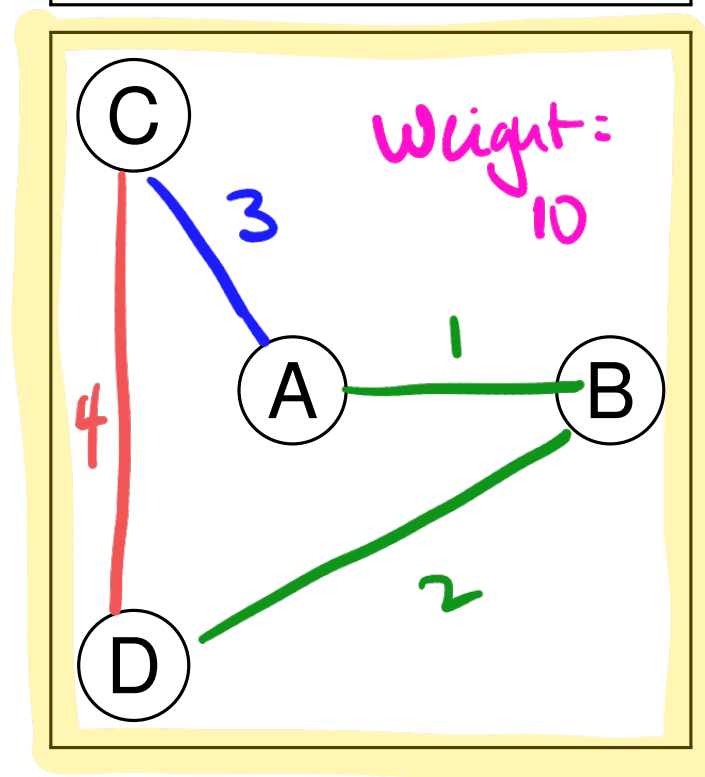
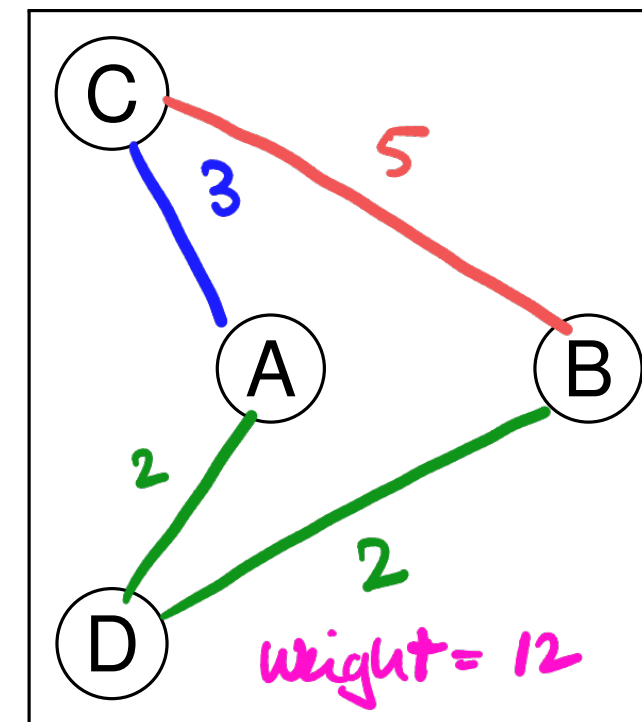
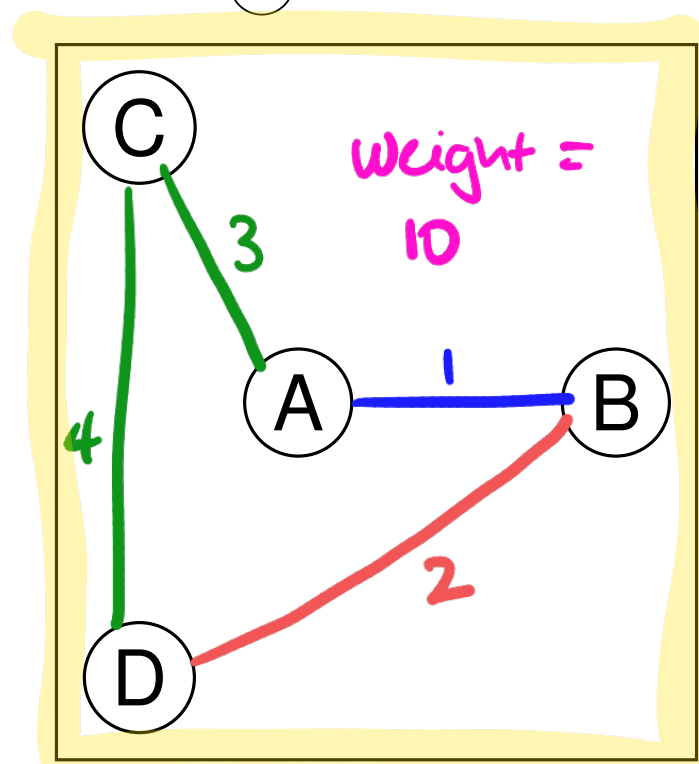
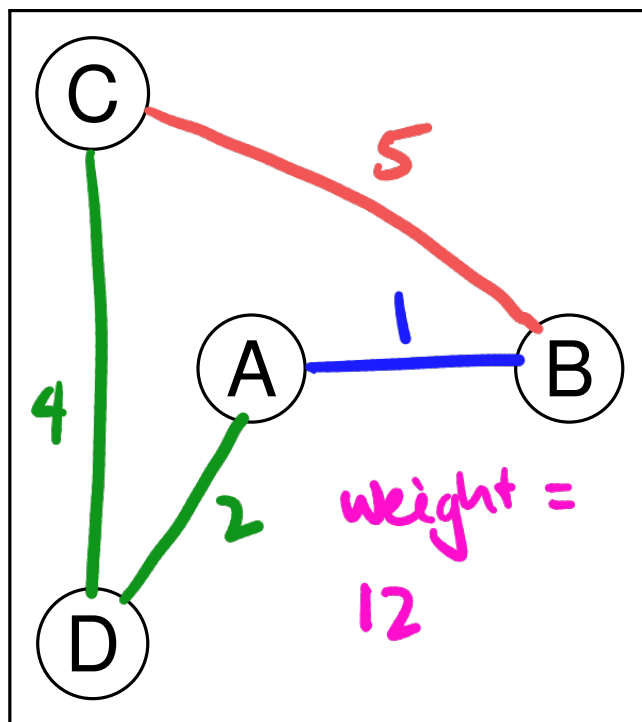
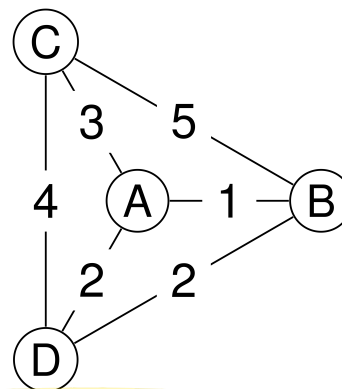


4 No circuit is possible



Can we find a Hamiltonian Circuit of smallest weight?

Brute Force Algorithm



Nearest Neighbor Algorithm (NNA)

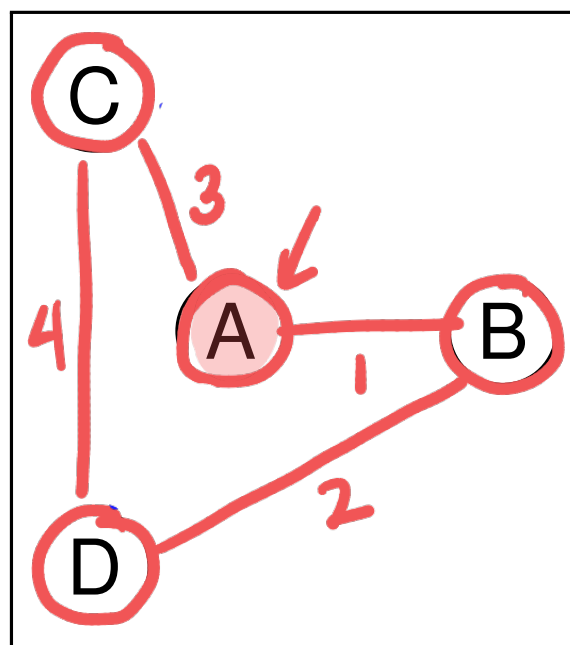
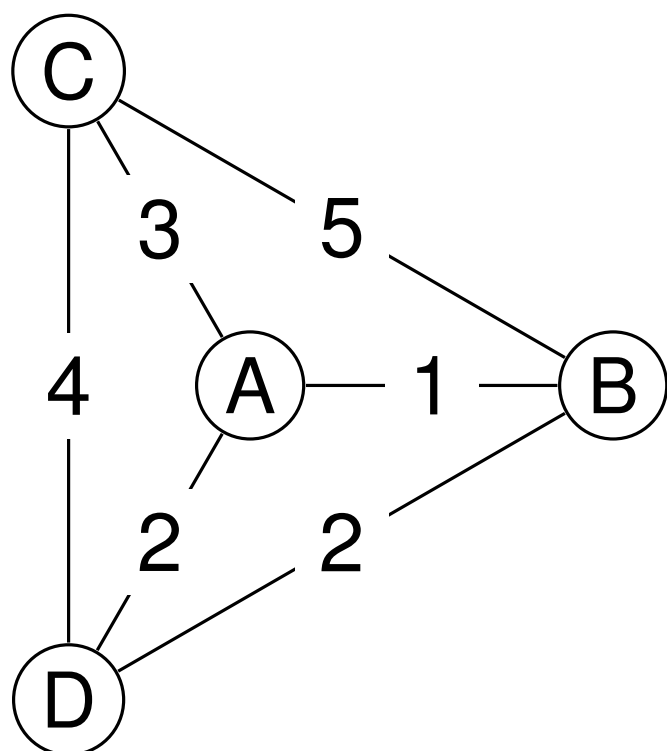
Choose a starting vertex

Go to the closest (by weight) vertex
as long as it doesn't close the circuit
too soon!

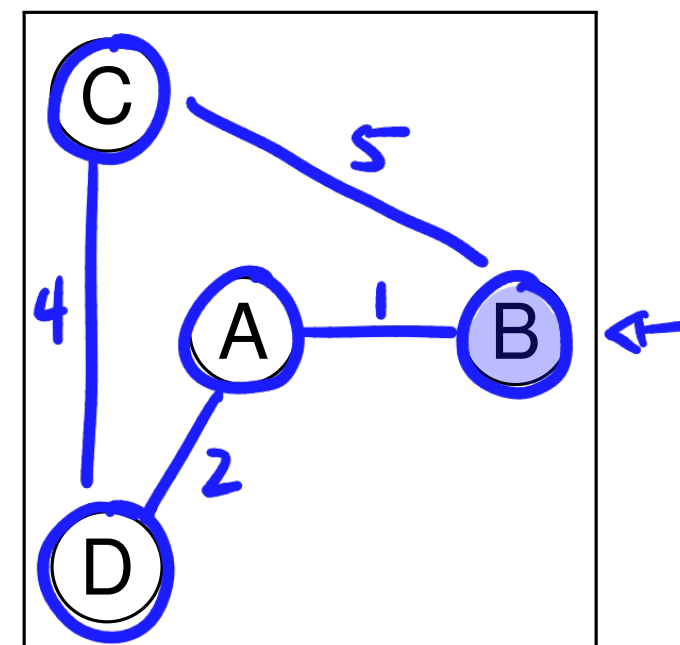
Problem: it does not guarantee you find
a minimal Hamiltonian circuit!

Nearest Neighbor Algorithm (NNA)

On the graph below, complete nearest neighbor starting with starting vertex A. What happens if you start at B?



$$\begin{aligned}\text{Weight} &= 4 + 3 + 2 + 1 \\ &= 10\end{aligned}$$



$$\begin{aligned}\text{Weight} &= 12 \\ \text{This is not minimal!}\end{aligned}$$

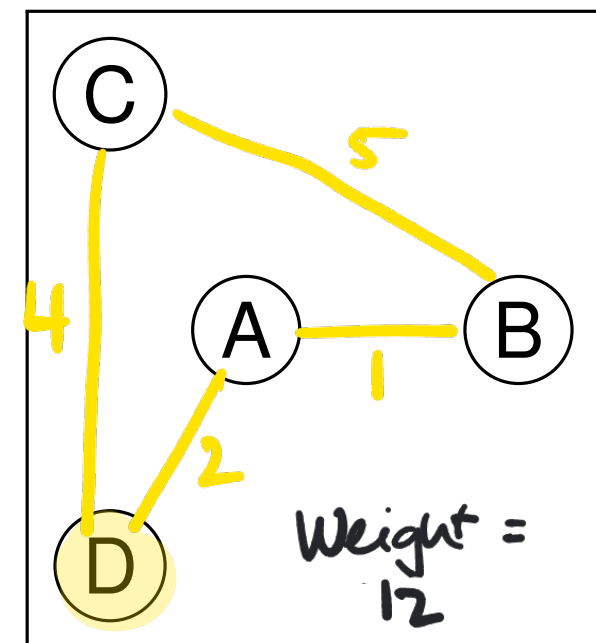
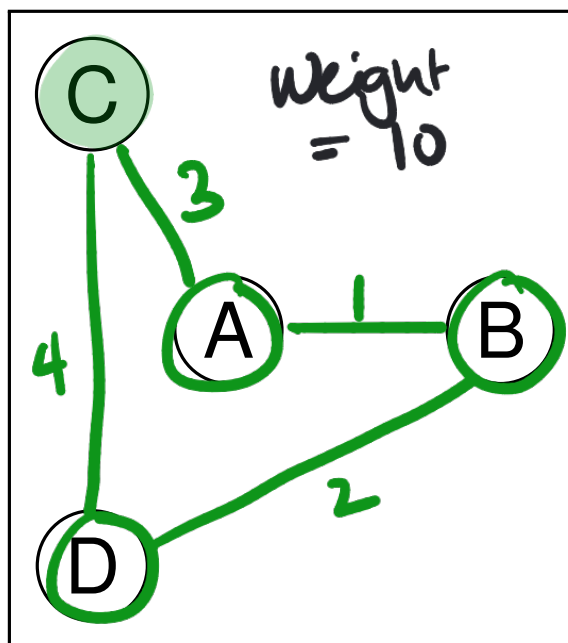
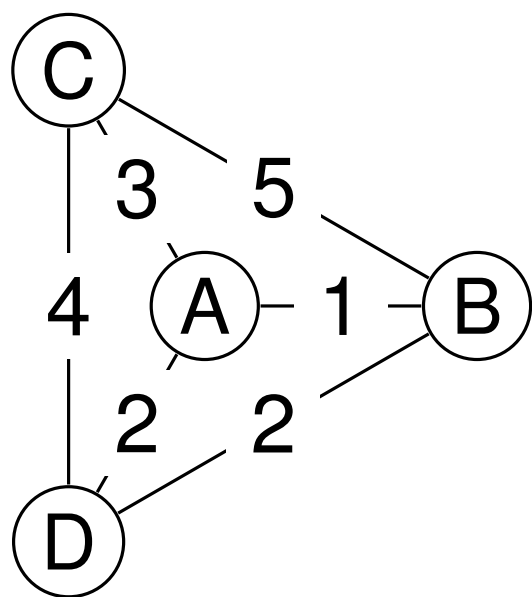
Repeated Nearest Neighbor Algorithm (RNNA)

Do NNA starting at every vertex
Choose the circuit among those that is
Smallest weight

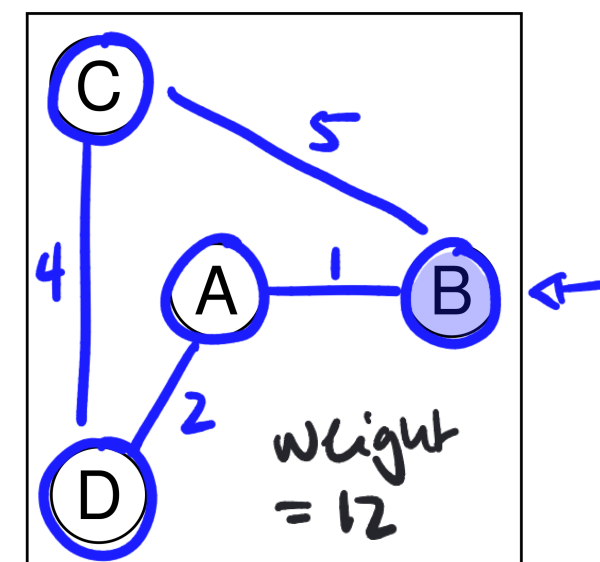
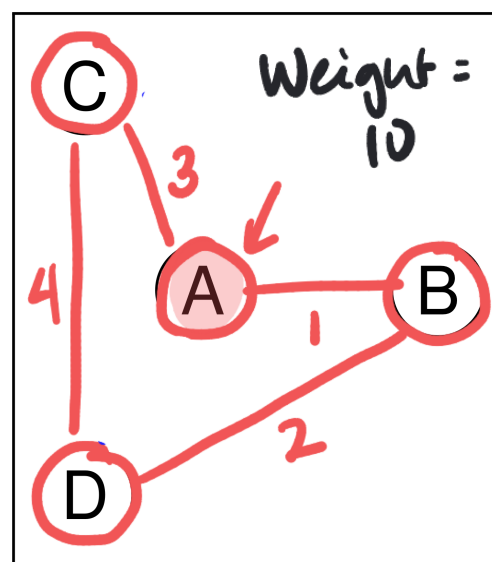
Bad News... this doesn't guarantee an optimal circuit either

Repeated Nearest Neighbor Algorithm (RNNA)

On the graph below, complete the Repeated Nearest Neighbor.

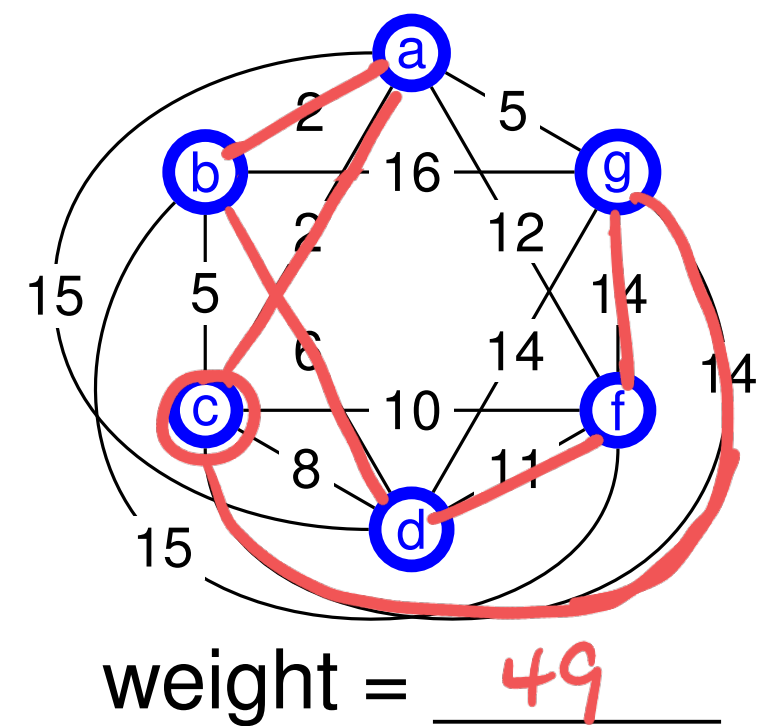
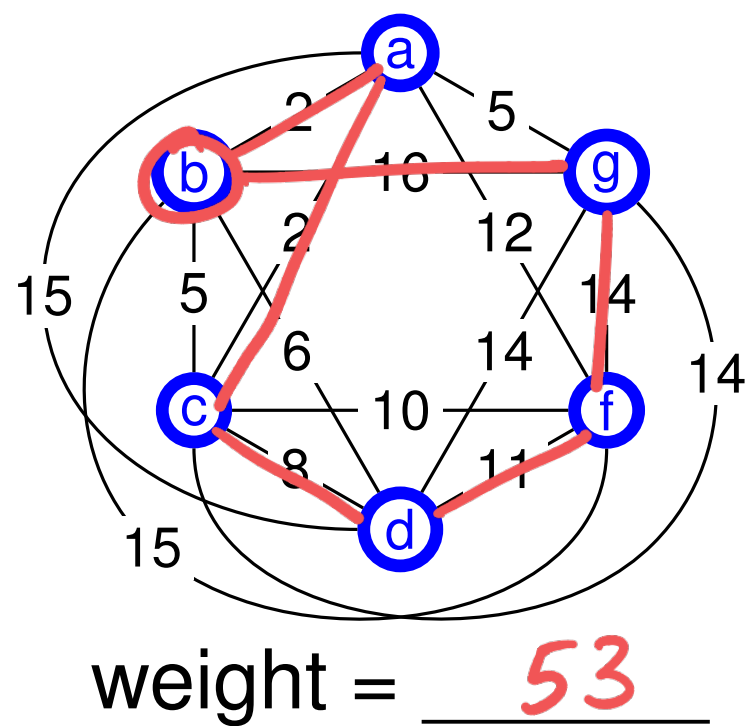
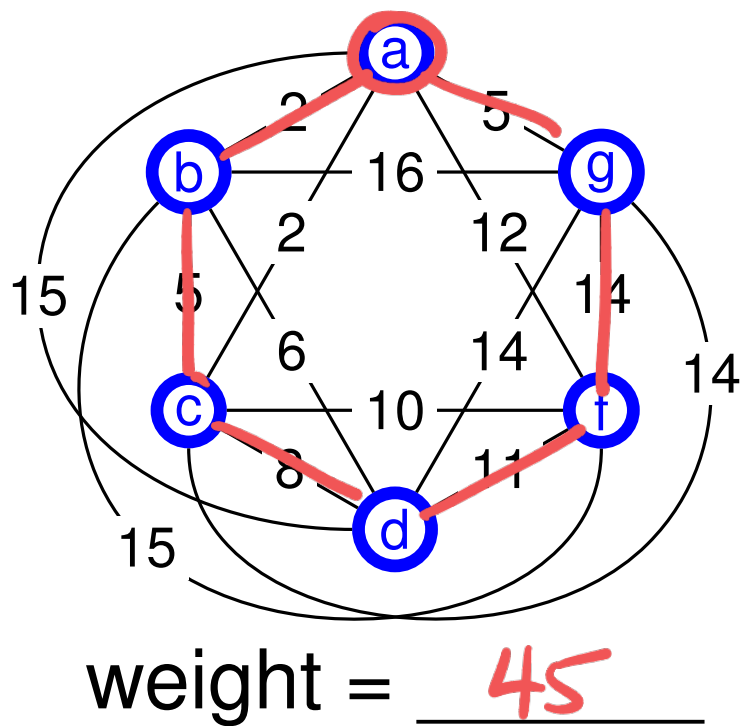


RNNA
returns a
circuit of weight
10

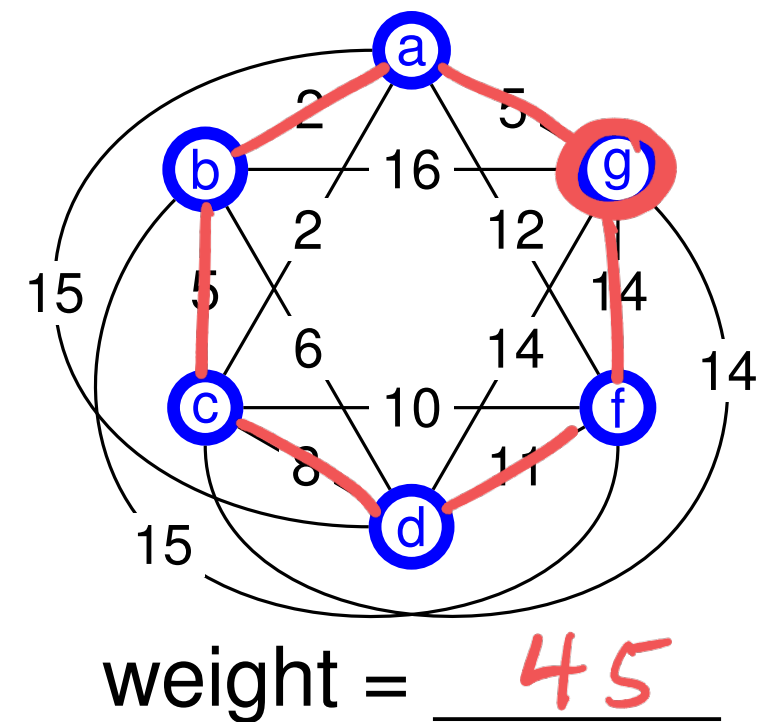
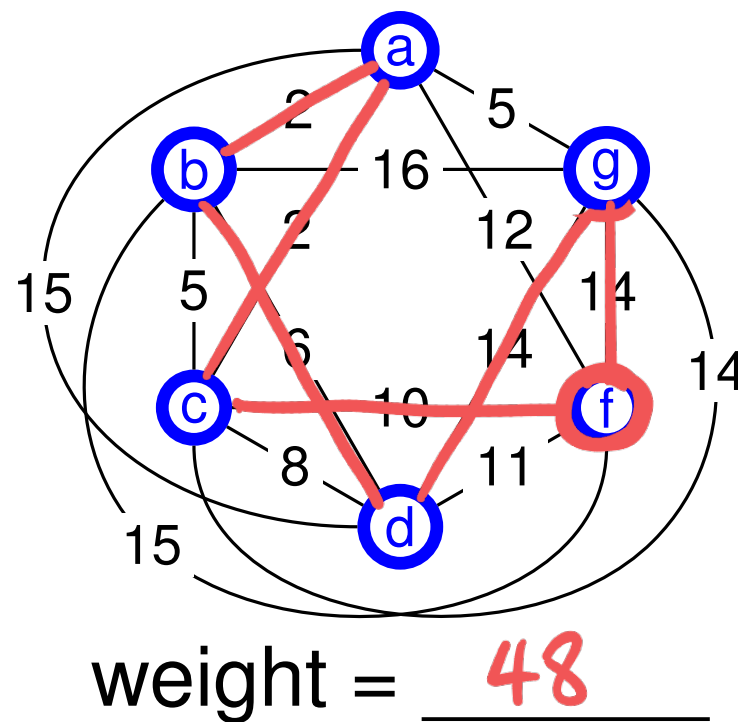
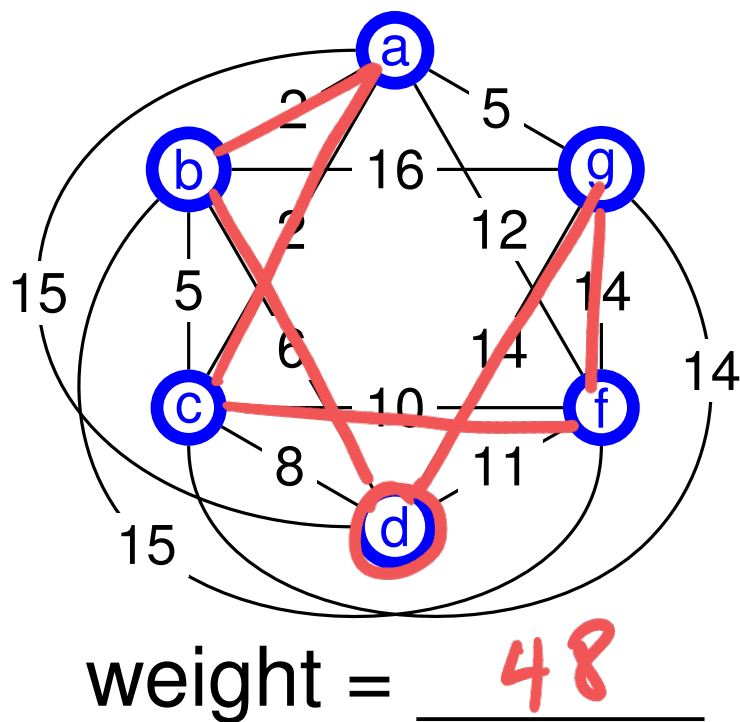


Complete Repeated Nearest Neighbor

Complete Repeated Nearest Neighbor on the graph below. (Suggestion: Use parallel processing)



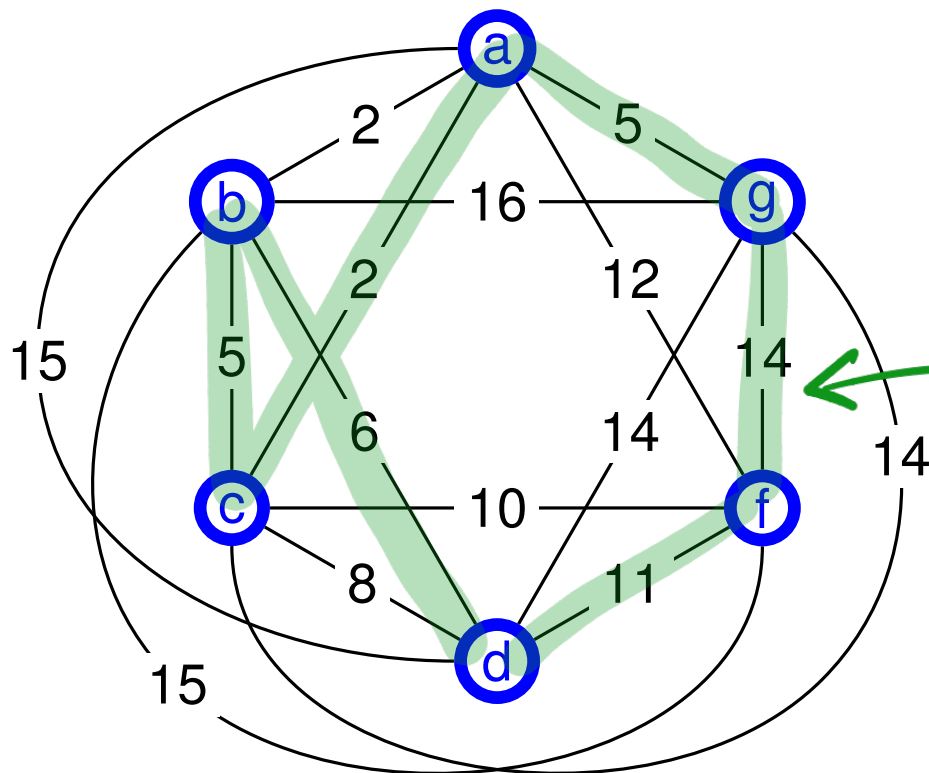
Complete Repeated Nearest Neighbor



Conclusion:

Weights:

a 45 b 53 c 49 d 48 f 48 g 45



Best circuit from RNA
is weight 45

a b c d f g

Total weight

$$= 5 + 2 + 5 + 6 + 11 + 14$$